Using Conda and Jupyter Lab

Assumption: Miniforge is already installed and initialized. **Windows:** use Miniforge Prompt. **macOS/Linux:** use Terminal.

1. Create and prepare a Conda environment

If you have not yet created an environment, replace `env_name` below with the name you want to use.

```
conda create -n env_name python
```

```
conda activate env_name
```

```
conda install pandas pillow numpy matplotlib jupyter ipykernel
```

```
python -m ipykernel install --user --name=env_name
```

After activation, the environment name should be visible in parentheses on the left of the prompt, for example `(env_name)`. The name used with `--name=` must be the same as your Conda environment.

Example: if the environment is called `bioimaging`, use:

```
conda create -n bioimaging python
conda activate bioimaging
python -m ipykernel install --user --name=bioimaging
```

2. Open the notebooks in Jupyter Lab

Navigate to the directory containing the notebooks:

```
cd <directory>
```

```
cd ~/Documents/Introduction_BioImaging_Course/Day5
```

Make sure your Conda environment is still activated and visible as `(env_name)`. Then start Jupyter Lab:

```
jupyter lab
```

Jupyter Lab should open in a new tab in your default browser.

1. Find the notebook in the file browser on the left.

2. Double-click the notebook to open it.

3. Look in the **top-right corner** of the notebook.

4. Click `Python 3 (ipykernel)` or `No kernel`.

5. Select the environment you created earlier (for example `bioimaging`).

Done: the notebook will now run using the Python packages installed in that Conda environment.

3. Quit Jupyter and install additional packages

To quit Jupyter Lab, go back to the terminal or Miniforge Prompt that you used to launch it and press **Ctrl-C**. If asked to confirm shutdown, type `y` and press Enter.

If you want to install a new package later, first close Jupyter Lab. Make sure your Conda environment is still active, install the package, and then relaunch Jupyter Lab.

```
conda install package_name
jupyter lab
```